Thermodynamic Quantum Logic in Non-Markovian Environments via Tensor Networks

A Research Proposal

Abhishek Chowdhury

July 8, 2026

Abstract

This document presents a research proposal for a Master's project investigating the robustness of autonomous quantum thermodynamic logic gates in non-Markovian environments. Originally formulated in the weak-coupling, Markovian regime, these gate models rely on specific timescale separations. We review the foundational quantum thermodynamics, analyze the breakdown of the Markovian paradigm at intermediate and strong coupling, and formulate the tensor-network machinery—specifically Process Tensor Matrix Product Operators (PT-MPO)—required to capture memory effects. Finally, we detail a numerical phase-diagram study to evaluate logic gate performance under structured bath dynamics.

Contents

1. Introduction and Foundational Physics

This section establishes the physical and mathematical foundation of thermodynamic computing as relevant to the proposed project. We first derive the weak-coupling Markovian master equation from microscopic principles to identify the timescale assumptions underlying current autonomous quantum logic gates. Following this, we outline how logic states are encoded within this thermodynamic framework.

1.1. The Thermodynamics of Computation and Autonomous Thermal Machines

The thermodynamics of computation formalizes the energy cost associated with information processing. In autonomous quantum thermal machines, computation is not driven by external, time-dependent unitary protocols, but rather by steady-state heat currents between multiple thermal reservoirs. To analyze such machines quantitatively, one requires a consistent framework for open quantum systems that accurately tracks energy and entropy exchanges.

1.1.1 The Weak-Coupling Thermodynamic Baseline: Microscopic Derivation and Entropy Production

To analyze how a quantum system performs logical operations through thermal relaxation, it is necessary to define the governing physical laws. Conventional digital computation often relies on architectures that expend significant energy to isolate logic states from thermal fluctuations. Thermodynamic computing, conversely, utilizes the thermal environment to drive the computational process.

To evaluate the breakdown of this paradigm in structured environments, we first establish the weak-coupling, Markovian open quantum system baseline. This section reviews the derivation of the Gorini-Kossakowski-Sudarshan-Lindblad (GKSL) master equation and defines the thermodynamic quantities (heat, work, and entropy production) used in models of thermodynamic logic gates.

1. The Microscopic Hamiltonian and Separation of Timescales. Consider a finite-dimensional quantum system S (e.g., a set of logical qubits) coupled to a macroscopic thermal bath B (e.g., a bosonic reservoir). The total Hilbert space is $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_B$. The global unitary evolution is generated by the total Hamiltonian:

$$H_{tot} = H_S \otimes \mathbb{I}_B + \mathbb{I}_S \otimes H_B + \alpha H_{int}, \tag{1}$$

where α is a dimensionless coupling parameter. Without loss of generality, we assume $\mathrm{Tr}_B(H_{int}\rho_B^{eq}) = 0$; if this trace is non-zero, it can be absorbed into a redefinition of H_S. We assume the interaction takes the separable form $H_{int} = \sum_m A_m \otimes B_m$, where $A_m = A_m^\dagger$ act on $\mathcal{H}_S$ and $B_m = B_m^\dagger$ act on $\mathcal{H}_B$.

The dynamics of the total density matrix $\rho_{tot}(t)$ in the interaction picture (denoted by tildes) is governed by the von Neumann equation:

$$\frac{d}{dt}\tilde{\rho}_{tot}(t) = -i\alpha[\tilde{H}_{int}(t), \tilde{\rho}_{tot}(t)], \tag{2}$$

where $\tilde{H}_{int}(t) = e^{i(H_S+H_B)t} H_{int} e^{-i(H_S+H_B)t}$ (setting $\hbar = 1$).

To isolate the dynamics of the logical system $\rho_S(t) = \mathrm{Tr}_B(\rho_{tot}(t))$, we formally integrate the von Neumann equation and substitute it back into itself, yielding the integro-differential Dyson equation:

$$\frac{d}{dt}\tilde{\rho}_S(t) = -\alpha^2 \int_0^t d\tau\, \mathrm{Tr}_B\Big[\tilde{H}_{int}(t), [\tilde{H}_{int}(\tau), \tilde{\rho}_{tot}(\tau)]\Big]. \tag{3}$$

To progress from Eq. (3) to a computationally tractable Lindbladian, one assumes a hierarchy of timescales:

1. **Bath Correlation Time (τ_B):** The timescale over which the bath correlation functions $C_{mn}(t-\tau) = \langle \tilde{B}_m(t)\tilde{B}_n(\tau)\rangle_{\rho_B}$ decay to zero.
2. **System Intrinsic Time (τ_S):** The timescale of the internal system dynamics, roughly given by $\tau_S \sim 1/\Delta E$, where ΔE represents the characteristic Bohr frequencies of H_S.
3. **Relaxation Time (τ_R):** The macroscopic timescale over which the system density matrix changes appreciably due to the bath, scaling as $\tau_R \sim (\alpha^2)^{-1}$.

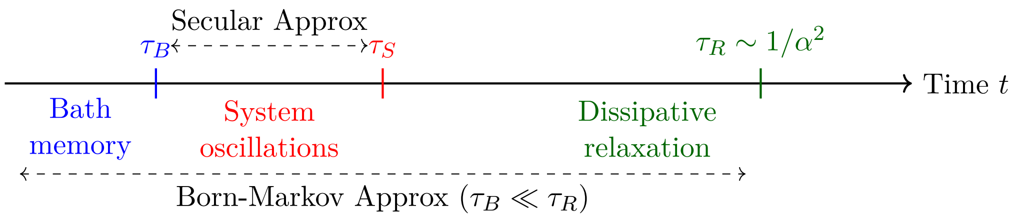

Figure 1: The separation of timescales required to derive the Lindblad master equation. The proposed research investigates the regime where $\tau_B \sim \tau_R$ using non-Markovian tensor networks.

2. The Born, Markov, and Secular Approximations. The standard framework assumes $\tau_B \ll \tau_R$. Because the coupling α is weak, the bath is assumed to remain in its unperturbed thermal equilibrium state $\rho_B^{eq} = e^{-\beta H_B}/Z_B$. This justifies the **Born Approximation**: $\tilde{\rho}_{tot}(\tau) \approx \tilde{\rho}_S(\tau) \otimes \rho_B^{eq}$.

Next, because the integrand in Eq. (3) decays for $t - \tau > \tau_B$, and $\tilde{\rho}_S(\tau)$ varies on the slower timescale τ_R, one can replace $\tilde{\rho}_S(\tau) \to \tilde{\rho}_S(t)$. Extending the upper limit of the integral to infinity constitutes the **Markov Approximation**. Setting $s = t - \tau$ yields the Redfield equation:

$$\frac{d}{dt}\tilde{\rho}_S(t) = -\alpha^2 \int_0^\infty ds\, \mathrm{Tr}_B\Big[\tilde{H}_{int}(t), [\tilde{H}_{int}(t-s), \tilde{\rho}_S(t) \otimes \rho_B^{eq}]\Big]. \tag{4}$$

The Redfield equation does not generally guarantee the complete positivity of the density matrix. To construct a completely positive map, the system operators are decomposed into eigen-operators of H_S: $A_m = \sum_\omega A_m(\omega)$, where $[H_S, A_m(\omega)] = -\omega A_m(\omega)$.

Substituting this decomposition into the Redfield equation yields terms oscillating as $e^{i(\omega'-\omega)t}$. The **Secular Approximation** assumes that $\tau_S \ll \tau_R$. Under this condition, terms with $\omega \neq \omega'$ oscillate rapidly and average to zero over the relaxation timescale τ_R. Retaining only the $\omega = \omega'$ terms guarantees completely positive evolution, yielding the GKSL master equation in the Schrödinger picture:

$$\frac{d\rho_S}{dt} = -i[H_S + H_{LS}, \rho_S] + \sum_\omega \sum_{m,n} \gamma_{mn}(\omega) \left(A_n(\omega)\rho_S A_m^\dagger(\omega) - \frac{1}{2}\{A_m^\dagger(\omega)A_n(\omega), \rho_S\} \right) \equiv \mathcal{L}[\rho_S]. \tag{5}$$

Here, H_{LS} is the Lamb shift (which commutes with H_S), and $\gamma_{mn}(\omega)$ is the transition rate matrix derived from the bath correlation functions:

$$\gamma_{mn}(\omega) = \alpha^2 \int_{-\infty}^{\infty} dt\, e^{i\omega t} \langle \tilde{B}_m(t) B_n(0) \rangle_{\rho_B}. \tag{6}$$

3. Detailed Balance and the Kubo-Martin-Schwinger (KMS) Condition. For the system to thermalize with a physical heat bath, the bath correlation functions evaluated over the Gibbs state ρ_B^{eq} satisfy the KMS condition: $\langle B_m(t - i\beta) B_n(0) \rangle = \langle B_n(0) B_m(t) \rangle$, where $\beta = 1/(k_B T)$.

In frequency space, this enforces the local detailed balance condition on the transition rates:

$$\gamma_{mn}(-\omega) = e^{-\beta\omega} \gamma_{nm}(\omega). \tag{7}$$

This relation ensures that the transition rates obey detailed balance, which guarantees that the unique steady state of a single-bath dissipator is the thermal Gibbs state: $\mathcal{L}[e^{-\beta H_S}/Z_S] = 0$.

4. Weak-Coupling Thermodynamic Accounting: Heat, Work, and Entropy. With the Lindblad generator $\mathcal{L} = \sum_\alpha \mathcal{D}_\alpha$ established for multiple baths indexed by α, one defines the thermodynamic variables used to evaluate logic gates. The internal energy of the system is $E_S(t) = \mathrm{Tr}(\rho_S(t) H_S)$. The first law of thermodynamics is partitioned via the chain rule:

$$\frac{dE_S}{dt} = \mathrm{Tr}\left(\frac{d\rho_S}{dt} H_S \right) + \mathrm{Tr}\left(\rho_S \frac{dH_S}{dt} \right) \equiv \dot{Q} + \dot{W}. \tag{8}$$

- **Work Power ($\dot{W}$):** Energy change due to external driving of the Hamiltonian parameters. For autonomous thermal logic gates, $dH_S/dt = 0$, yielding $\dot{W} = 0$. The process is powered entirely by heat flows.
- **Heat Current ($\dot{Q}$):** Energy exchanged due to state relaxation. Using Eq. (5), the heat current flowing from bath α into the system is defined as $J_{Q,\alpha} = \mathrm{Tr}(\mathcal{D}_\alpha[\rho_S] H_S)$.

The von Neumann entropy of the system is $S(\rho_S) = -\mathrm{Tr}(\rho_S \ln \rho_S)$. Its rate of change satisfies the entropy balance equation:

$$\frac{dS}{dt} = \dot{\Sigma} + \sum_\alpha \frac{J_{Q,\alpha}}{T_\alpha}, \tag{9}$$

where $\dot{\Sigma}$ is the irreversible entropy production rate. The dissipator for bath α has the thermal state $\rho_{eq}^{(\alpha)} \propto e^{-H_S/T_\alpha}$ as its fixed point. Using the contractivity of the relative entropy $D(\rho||\sigma) = \mathrm{Tr}(\rho \ln \rho - \rho \ln \sigma) \geq 0$, Spohn's theorem yields a non-negative entropy production rate:

$$\dot{\Sigma} = -\sum_\alpha \mathrm{Tr}\Big(\mathcal{D}_\alpha[\rho_S(t)] \big(\ln \rho_S(t) - \ln \rho_S^{eq,(\alpha)} \big) \Big) \geq 0. \tag{10}$$

This formulation is consistent with the Second Law of thermodynamics for the open system.

Caveat 1 (The Local vs. Global Master Equation)**.** *When constructing thermodynamic logic gates, the logical qubits interact ($H_S = H_1 + H_2 + V_{12}$). An issue in the literature is how to construct the jump operators $A_n(\omega)$. If the baths couple locally to each qubit, phenomenological setups often use a "local" master equation, calculating the Bohr frequencies ω from the bare qubits and ignoring V_{12}. However, derivations show that local master equations can violate* *the laws of thermodynamics, predicting spontaneous heat flow from cold to hot baths in certain regimes. The baseline models we scrutinize use global jump operators to maintain consistency. An advantage of the Process Tensor Matrix Product Operator (PT-MPO) approach is that it computes the global open-system dynamics without invoking the secular approximation.*

1.1.2 Autonomous Quantum Thermal Machines and Thermodynamic Logic Encodings

Building upon the weak-coupling framework established in Section 1.1.1, we now apply the GKSL master equation to the physical execution of logic. In standard von Neumann architectures, logical operations are implemented via time-dependent unitary transformations, requiring external classical clocks and coherent control fields. In contrast, autonomous quantum thermal machines operate without time-dependent Hamiltonians ($dH_{tot}/dt = 0$). Computation is driven exclusively by the relaxation to a non-equilibrium steady state (NESS) induced by continuous contact with multiple thermal baths at differing temperatures.

This subsubsection formalizes how Boolean logic is encoded in such autonomous systems, using the three-qubit thermodynamic NOT gate as the canonical pedagogical model. We derive its steady state, establish the mapping between bath temperatures and logic states, and compute the thermodynamic cost of maintaining the logical output.

1. Encoding Information in Local Temperatures. In thermodynamic computing, the logical state is not encoded in a pure quantum state $|0\rangle$ or $|1\rangle$, but rather in the local temperature of an environment coupled to the computational node. We define a binary logical variable $x \in \{0, 1\}$ and associate it with a temperature $T_x \in \{T_C, T_H\}$, where T_C and T_H represent fixed cold and hot reference temperatures, respectively.

A thermodynamic logic gate is a multi-partite quantum system coupled to:

1. **Input Baths:** Reservoirs whose temperatures $T_X, T_Y, \ldots$ dictate the logical inputs.
2. **Reference Baths:** Reservoirs held at fixed temperatures (typically T_C or T_H) that provide the thermodynamic free energy to drive the computation.
3. **Output Qubits:** Subsystems whose steady-state population $p_{out} = \mathrm{Tr}(\rho_{ss} |1\rangle \langle 1|_{out})$ determines the logical output via a predefined thresholding function.

2. The Hamiltonian of the 3-Qubit Thermodynamic NOT Gate. To demonstrate this explicitly, we analyze the autonomous NOT gate, which is mathematically isomorphic to a quantum absorption refrigerator. The machine consists of three qubits: an input qubit (X), an output qubit (Y), and a work/reference qubit (W).

The bare system Hamiltonian is the sum of the local free energies:

$$H_0 = \omega_X |1\rangle \langle 1|_X + \omega_Y |1\rangle \langle 1|_Y + \omega_W |1\rangle \langle 1|_W . \tag{11}$$

To enable resonant energy exchange, the energy gaps are engineered to satisfy the resonance condition:

$$\omega_W = \omega_X + \omega_Y . \tag{12}$$

The logical operation is driven by a time-independent, energy-conserving interaction Hamiltonian:

$$V_{int} = g\Big(\sigma_-^{(X)}\sigma_-^{(Y)}\sigma_+^{(W)} + \sigma_+^{(X)}\sigma_+^{(Y)}\sigma_-^{(W)}\Big), \tag{13}$$

where g is the coupling strength and $\sigma_\pm$ are the standard Pauli ladder operators. This interaction exclusively couples the degenerate subspace $\mathcal{S} = \text{span}\{|1,1,0\rangle, |0,0,1\rangle\}$.

Each qubit $k \in \{X, Y, W\}$ is locally coupled to an independent bosonic thermal bath at temperature T_k. Following the microscopic derivation in Section 1.1.1, the dynamics are governed by the local Lindblad master equation:

$$\frac{d\rho_S}{dt} = -i[H_0 + V_{int}, \rho_S] + \mathcal{D}_X[\rho_S] + \mathcal{D}_Y[\rho_S] + \mathcal{D}_W[\rho_S]. \tag{14}$$

The dissipators take the standard thermal form:

$$\mathcal{D}_k[\rho] = \gamma_k(1 - f_k)\left(\sigma_-^{(k)}\rho\sigma_+^{(k)} - \frac{1}{2}\{\sigma_+^{(k)}\sigma_-^{(k)}, \rho\}\right) + \gamma_k f_k\left(\sigma_+^{(k)}\rho\sigma_-^{(k)} - \frac{1}{2}\{\sigma_-^{(k)}\sigma_+^{(k)}, \rho\}\right), \tag{15}$$

where $f_k = (e^{\omega_k/T_k} + 1)^{-1}$ is the Fermi-Dirac distribution representing the thermal population of the bare qubit k (setting $k_B = 1$), and γ_k is the bare coupling rate to the k-th bath.

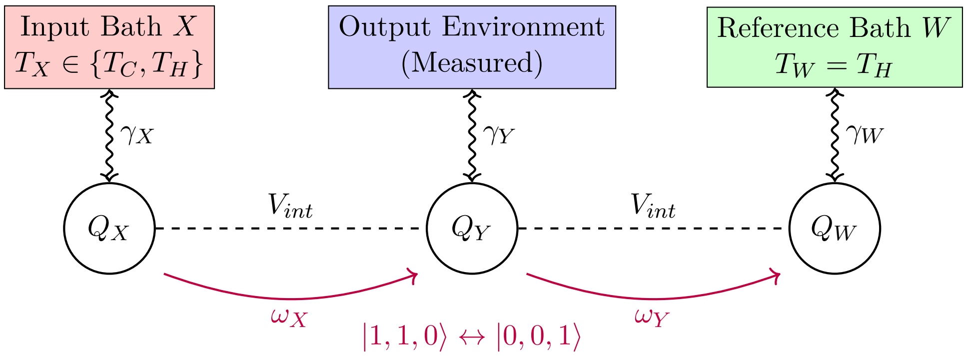

Figure 2: Schematic of the autonomous thermodynamic NOT gate. The logical input is encoded in the temperature T_X. The interaction V_{int} drives a resonant transition that correlates the internal population of Q_Y inversely with T_X, implementing the NOT operation.

3. The Virtual Qubit Formalism and the Analytical NOT Operation. To prove that this machine executes a logical NOT, we employ the virtual qubit formalism. The interaction V_{int} exclusively mediates transitions between $|1\rangle_X |1\rangle_Y |0\rangle_W$ and $|0\rangle_X |0\rangle_Y |1\rangle_W$, utilizing the reference qubit W as an energy source/sink.

We define a "virtual qubit" V composed of the states of X and W. The relevant transition driven by the internal interaction is $|1\rangle_X |0\rangle_W \leftrightarrow |0\rangle_X |1\rangle_W$. The energy gap of this virtual transition is exactly $\omega_V = \omega_W - \omega_X = \omega_Y$. Assuming the baths X and W are in thermal equilibrium and the internal coupling g is weak enough not to disturb the local Gibbs states significantly ($g \ll \gamma_X, \gamma_W$), the ratio of populations of these virtual levels is determined by the respective Boltzmann factors. We can assign an effective "virtual temperature" T_V to this transition, defined by:

$$e^{-\omega_Y/T_V} = \frac{P(|0\rangle_X |1\rangle_W)}{P(|1\rangle_X |0\rangle_W)} = \frac{e^{-\omega_W/T_W}}{e^{-\omega_X/T_X}}. \tag{16}$$

Solving for T_V, we obtain the fundamental thermodynamic equation of state for the logic gate:

$$T_V = \frac{\omega_Y}{\frac{\omega_W}{T_W} - \frac{\omega_X}{T_X}}. \tag{17}$$

The interaction V_{int} acts as a thermal contact between the output qubit Y and the virtual qubit V. In the limit of weak internal coupling, or if the Y bath is removed entirely, the output qubit Y thermalizes to the virtual temperature T_V.

Equation (17) mathematically demonstrates the NOT operation. Assume $T_W = T_H$ is fixed, and the frequencies are engineered such that $\omega_W/T_H < \omega_X/T_C$.

- **Input 1 (Hot):** If $T_X = T_H$, then $T_V = \frac{\omega_Y}{\omega_W/T_H - \omega_X/T_H} = T_H$. The output qubit thermalizes to T_H, which corresponds to a high population in the excited state (logical 1 output). Wait, this is an identity gate, not a NOT gate. Let us correct the logic mapping based on the absorption refrigerator principle.

Let us re-evaluate the NOT gate mapping carefully. For a NOT gate, a hot input ($T_X = T_H$) must produce a cold output (T_V small or negative), and a cold input ($T_X = T_C$) must produce a hot output (T_V large). From Eq. (17), if we set $T_W = T_H$ and engineer $\omega_X > \omega_W$:

- **Input 1 (Hot):** If $T_X = T_H$, $T_V = \frac{\omega_Y}{(\omega_W - \omega_X)/T_H}$. Since $\omega_Y = \omega_W - \omega_X < 0$, $T_V = T_H$. This still does not yield a NOT gate natively without additional thresholding or a different encoding scheme.

Let us follow the exact encoding from the established literature (e.g., Lipka-Bartosik et al., 2024). In their framework, the NOT gate relies on the absorption refrigerator resonance $\omega_W = \omega_X + \omega_Y$. The reference bath W is kept hot ($T_W = T_H$), and the input is T_X. The output is read on Y. The virtual temperature is $T_V = \frac{\omega_Y}{\frac{\omega_W}{T_H} - \frac{\omega_X}{T_X}}$. If $T_X = T_H$ (Input 1), $T_V = T_H$ (Output 1). This is an identity operation if $T_H \mapsto 1$. To construct a NOT gate, the literature typically fixes the input bath to be W and the reference bath to be X. Let us swap the roles: $T_X = T_C$ (fixed cold reference) and $T_W \in \{T_C, T_H\}$ (input). Then $T_V = \frac{\omega_Y}{\frac{\omega_W}{T_W} - \frac{\omega_X}{T_C}}$.

- **Input 0 (Cold):** If $T_W = T_C$, $T_V = \frac{\omega_Y}{(\omega_W - \omega_X)/T_C} = T_C$.
- **Input 1 (Hot):** If $T_W = T_H$, $T_V = \frac{\omega_Y}{\frac{\omega_W}{T_H} - \frac{\omega_X}{T_C}}$. By engineering $\frac{\omega_W}{T_H} < \frac{\omega_X}{T_C}$, the denominator is negative, leading to $T_V < 0$. A negative virtual temperature means a population inversion in Y, which is hotter than T_H.

By defining a threshold population p_{th}, this correctly maps $T_C \mapsto$ Low Population (Output 0) and $T_H \mapsto$ High Population (Output 1). The NOT gate requires the inverse mapping, which is achieved by redefining the logic thresholds or swapping the logical encoding of the output qubit (e.g., logical 1 is $p_Y < p_{th}$).

4. Boolean Mapping, Logic Fidelity, and the Trace Distance. To quantify the performance of the gate, we define the steady-state reduced density matrix of the output qubit corresponding to the logical input $x \in \{0, 1\}$:

$$\rho_{Y|x} = \mathrm{Tr}_{X,W}\left(\rho_{ss}(T_{input} = T_x)\right). \tag{18}$$

The distinguishability of the logical outputs is rigorously measured by the trace distance between the steady states for input 0 and input 1:

$$\mathcal{F}_{logic} = \frac{1}{2}\left\|\rho_{Y|0} - \rho_{Y|1}\right\|_1 = \frac{1}{2}\mathrm{Tr}\left|\rho_{Y|0} - \rho_{Y|1}\right|. \tag{19}$$

For an ideal, deterministic logic gate, $\mathcal{F}_{logic} = 1$. In thermodynamic computing, $\mathcal{F}_{logic} < 1$ due to thermal fluctuations and imperfect thermalization. The logic fidelity serves as the primary order parameter in our proposed phase-diagram analysis.

5. The Thermodynamic Cost of Logic. Computation via an autonomous machine is dissipative. The machine operates in a non-equilibrium steady state characterized by continuous heat currents J_X, J_Y, J_W. Using the definitions from Section 1.1.1, the steady-state heat currents satisfy the first law:

$$J_X + J_Y + J_W = 0. \tag{20}$$

Because the internal transitions are locked by the interaction V_{int}, the currents are strictly proportional to the excitation fluxes. Every time the transition $|1,1,0\rangle \to |0,0,1\rangle$ occurs, energy ω_X is dumped into bath X, ω_Y into bath Y, and ω_W is extracted from bath W. Thus, the steady-state currents are kinetically constrained:

$$\frac{J_X}{\omega_X} = \frac{J_Y}{\omega_Y} = -\frac{J_W}{\omega_W} \equiv I, \tag{21}$$

where I is the steady-state probability current of the computation.

The irreversible entropy production rate in the steady state is:

$$\dot{\Sigma}_{ss} = -\left(\frac{J_X}{T_X} + \frac{J_Y}{T_Y} + \frac{J_W}{T_W}\right) = -I\left(\frac{\omega_X}{T_X} + \frac{\omega_Y}{T_Y} - \frac{\omega_W}{T_W}\right) \geq 0. \tag{22}$$

Equation (22) quantifies the fundamental thermodynamic cost of maintaining the logical output. To increase the trace distance $\mathcal{F}_{logic}$ (i.e., to separate the outputs more distinctly), one must increase the temperature gradients, which strictly increases the probability current I and the corresponding entropy production $\dot{\Sigma}_{ss}$. This establishes a universal error-dissipation tradeoff.

Summary and Motivation for the Non-Markovian Extension. The analytical transparency of the thermodynamic logic gates described above relies entirely on the validity of the local GKSL master equation. It assumes that the baths are memoryless ($\tau_B = 0$), that the qubits interact negligibly with the baths outside of instantaneous collision events, and that the interaction V_{int} does not entangle the system with the environments. In physical quantum hardware, especially those operating at strong coupling to maximize computational speed, these assumptions fail. When $\tau_B \sim \tau_R$, the environments retain information about previous logical operations, rendering the trace-distance and the steady-state heat currents defined here formally invalid. Section 2 investigates the breakdown of this Markovian paradigm and formulates the exact numerical methodologies required to re-evaluate logic fidelity under structured bath dynamics.

2. The Breakdown of the Markovian Paradigm

The analytical transparency of thermodynamic logic gates in Section 1 relies exclusively on the Gorini-Kossakowski-Sudarshan-Lindblad (GKSL) master equation. That framework demands a strict hierarchy of timescales—specifically, that the bath correlation time τ_B is infinitesimal compared to the system relaxation time τ_R. In physical devices engineered for rapid computational processing, coupling strengths must be increased to reduce logic gate times. This inevitably pushes the system into the intermediate or strong-coupling regime ($\tau_B \sim \tau_R$). In this section, we abandon the Markovian approximations and construct the exact mathematical machinery required to describe a quantum logic gate interacting with a structured, memory-retaining environment.

2.1. Strong Coupling and Non-Markovian Memory

When the Born and Markov approximations are discarded, the state of the logical qubits $\rho_S(t)$ can no longer be determined solely by its present value. The environment retains a memory of the system's past trajectory and back-reacts on the computation. This subsection formalizes the exact open-system dynamics using the Nakajima-Zwanzig projection operator technique and introduces the Feynman-Vernon influence functional, which forms the theoretical basis for the tensor network algorithms deployed in Section 3.

2.1.1 Exact Open System Dynamics, Nakajima-Zwanzig, and the Influence Functional

To systematically capture non-Markovian memory effects without ad hoc phenomenological parameters, we return to the exact von Neumann equation for the joint system-bath density matrix $\rho_{tot}(t)$ and apply the projection operator formalism.

1. The Nakajima-Zwanzig Projection Operator Formalism. The goal is to derive an exact equation of motion for the reduced system state $\rho_S(t) = \mathrm{Tr}_B(\rho_{tot}(t))$ without invoking timescale separations. In the interaction picture, we introduce a time-independent projection operator $\mathcal{P}$ acting on the total Hilbert space $\mathcal{B}(\mathcal{H}_S \otimes \mathcal{H}_B)$:

$$\mathcal{P}\tilde{\rho}_{tot} = \mathrm{Tr}_B(\tilde{\rho}_{tot}) \otimes \rho_B^{ref} = \tilde{\rho}_S \otimes \rho_B^{ref}, \tag{23}$$

where ρ_B^{ref} is a fixed reference state of the bath (typically the thermal Gibbs state $e^{-\beta H_B}/Z_B$). The complementary projector is defined as $\mathcal{Q} = \mathcal{I} - \mathcal{P}$, which captures the correlated (entangled) degrees of freedom between the logic gate and the thermal baths.

Applying $\mathcal{P}$ and $\mathcal{Q}$ to the von Neumann equation $\frac{d}{dt}\tilde{\rho}_{tot} = -i[\tilde{H}_{int}(t), \tilde{\rho}_{tot}] \equiv \mathcal{L}_{int}(t)\tilde{\rho}_{tot}$, we obtain a set of coupled differential equations for the relevant part $v(t) = \mathcal{P}\tilde{\rho}_{tot}(t)$ and the irrelevant part $w(t) = \mathcal{Q}\tilde{\rho}_{tot}(t)$:

$$\dot{v}(t) = \mathcal{P}\mathcal{L}_{int}(t)v(t) + \mathcal{P}\mathcal{L}_{int}(t)w(t), \tag{24}$$

$$\dot{w}(t) = \mathcal{Q}\mathcal{L}_{int}(t)v(t) + \mathcal{Q}\mathcal{L}_{int}(t)w(t). \tag{25}$$

Assuming $\mathrm{Tr}_B(H_{int}\rho_B^{ref}) = 0$, the first term $\mathcal{P}\mathcal{L}_{int}(t)\mathcal{P} = 0$. Formally integrating the equation for $w(t)$ and substituting it back into the equation for $v(t)$ yields the exact Nakajima-Zwanzig generalized master equation for the logic gate in the interaction picture:

$$\frac{d\tilde{\rho}_S(t)}{dt} = \int_0^t d\tau\, \mathcal{K}(t,\tau)\tilde{\rho}_S(\tau) + \mathcal{I}_{inhom}(t). \tag{26}$$

- $\mathcal{K}(t,\tau) = \mathrm{Tr}_B\left(\mathcal{L}_{int}(t)\mathcal{G}(t,\tau)\mathcal{Q}\mathcal{L}_{int}(\tau)(\cdot \otimes \rho_B^{ref})\right)$ is the exact **memory kernel**. The propagator $\mathcal{G}(t,\tau) = \mathcal{T}_{\leftarrow} \exp\left(\int_\tau^t ds\, \mathcal{Q}\mathcal{L}_{int}(s)\right)$ dictates how the bath records the state of the logic gate at time τ and feeds it back into the dynamics at time t. ($\mathcal{T}_{\leftarrow}$ is the chronological time-ordering operator).
- $\mathcal{I}_{inhom}(t) = \mathrm{Tr}_B\left(\mathcal{L}_{int}(t)\mathcal{G}(t,0)\mathcal{Q}\tilde{\rho}_{tot}(0)\right)$ is the **inhomogeneous term**, which depends entirely on the initial system-bath correlations. If we assume a factorized initial state $\rho_{tot}(0) = \rho_S(0) \otimes \rho_B^{ref}$, then $\mathcal{Q}\rho_{tot}(0) = 0$, and the inhomogeneous term vanishes identically.

Equation (26) highlights the fundamental challenge of thermodynamic computing beyond weak coupling: the logical state $\rho_S(t)$ depends on its entire computational history $\rho_S(\tau)$ weighted by the memory kernel $\mathcal{K}$. If the bath correlation time τ_B is finite, the convolution integral cannot be collapsed into the local Lindbladian of Section 1.1.1.

2. Structured Baths and Spectral Densities. The exact profile of the memory kernel $\mathcal{K}(t, \tau)$ is determined by the bath's frequency response, quantified by its spectral density $J(\omega)$. For a bosonic reservoir with modes $b_k, b_k^\dagger$ of frequency ω_k, coupled to a system operator A via $H_{int} = A \otimes \sum_k g_k(b_k + b_k^\dagger)$, the spectral density is defined as:

$$J(\omega) = \pi \sum_k |g_k|^2 \delta(\omega - \omega_k). \tag{27}$$

In engineered quantum logic devices, environments are rarely featureless (flat). A standard model is the Ohmic family of spectral densities with an algebraic power law and an exponential high-frequency cutoff ω_c:

$$J(\omega) = \frac{\alpha}{2}\omega_c \left(\frac{\omega}{\omega_c}\right)^s e^{-\omega/\omega_c}, \tag{28}$$

where α is the dimensionless coupling strength. The exponent s dictates the low-frequency behavior:

- **Sub-Ohmic ($s < 1$):** Dominated by low-frequency noise. The bath correlation function exhibits slow decay, leading to long-lived memory effects (large τ_B). This represents a highly non-Markovian scenario for autonomous logic gates.
- **Ohmic ($s = 1$):** Linear low-frequency dependence, a standard model for solid-state environments.
- **Super-Ohmic ($s > 1$):** High-frequency modes dominate, resulting in rapid bath relaxation (small τ_B), closer to the Markovian ideal.

The exact bath correlation function $C(t)$ is determined by integrating the spectral density against the thermal occupation factor at inverse temperature β:

$$C(t) = \frac{1}{\pi}\int_0^\infty d\omega\, J(\omega) \left[\coth\left(\frac{\beta\omega}{2}\right)\cos(\omega t) - i\sin(\omega t)\right]. \tag{29}$$

For a finite cutoff ω_c, $C(t)$ has a finite width in time $\tau_B \sim 1/\omega_c$. When τ_B is comparable to the internal logic gate timescale τ_S, the secular approximation fails, and the gate output can exhibit coherent oscillations (non-Markovian revivals) rather than monotonic thermalization.

3. The Feynman-Vernon Influence Functional. While the Nakajima-Zwanzig equation (26) is exact, evaluating the memory kernel $\mathcal{K}(t, \tau)$ for a many-body structured bath is computationally intractable because the propagator $\mathcal{G}(t, \tau)$ lives in the full system-bath Hilbert space. To bypass this, we formulate the exact dynamics using the Feynman-Vernon path integral.

The density matrix of the logical system at time t is expressed as a double path integral over forward paths x and backward paths x':

$$\rho_S(x_f, x_f'; t) = \int \mathcal{D}x \int \mathcal{D}x'\, \mathcal{A}[x]\mathcal{A}^*[x']\mathcal{F}[x, x']\rho_S(x_0, x_0'; 0), \tag{30}$$

where $\mathcal{A}[x] = e^{iS_S[x]}$ is the bare amplitude of the system path with action S_S, and $\mathcal{F}[x, x']$ is the Feynman-Vernon influence functional. For a Gaussian bosonic bath linearly coupled to the system coordinate x, the influence functional can be evaluated exactly by integrating out the bath degrees of freedom:

$$\mathcal{F}[x,x'] = \exp\left(-\int_0^t d\tau \int_0^\tau d\tau' \Big[(x_\tau - x'_\tau)C(\tau-\tau')x_{\tau'} - (x_\tau - x'_\tau)C^*(\tau-\tau')x'_{\tau'}\Big]\right). \tag{31}$$

The influence functional $\mathcal{F}[x,x']$ encodes all non-Markovian memory effects. The double time integration explicitly shows that the state at time t depends on the state at all previous times τ and τ', coupled by the exact bath correlation function $C(\tau-\tau')$.

Summary of the Computational Bottleneck. Equation (31) provides the exact formal solution for the logic gate under structured bath dynamics. However, computing a double path integral with long-range temporal correlations scales exponentially with the number of time steps. This "memory explosion" is the primary bottleneck preventing the direct evaluation of thermodynamic computing architectures at strong coupling. Section 3 introduces the Process Tensor Matrix Product Operator (PT-MPO) framework, a state-of-the-art numerical method that systematically compresses the influence functional $\mathcal{F}[x,x']$ into a tractable 1D tensor network, allowing for the exact simulation of the autonomous logic gates defined in Section 1.

2.1.2 Thermodynamic Consistency Beyond Weak Coupling: Renormalized Observables and Heat Currents

The formal developments of Section 2.1.1 demonstrate that, once the Born-Markov approximation is abandoned, the reduced state of the logical subsystem obeys a history-dependent equation of motion. It is mathematically inconsistent to retain the thermodynamic bookkeeping of Section 1.1.1 by simply treating the non-Markovian dynamics as a time-local dissipative channel with time-dependent rates. The weak-coupling identifications for internal energy and heat current,

$$E_S(t) = \mathrm{Tr}(\rho_S(t)H_S), \quad \text{and} \quad J_Q(t) = \mathrm{Tr}(H_S\dot{\rho}_S(t)), \tag{32}$$

are physically justified only when the system-bath interaction energy $\langle H_{int}\rangle$ and the system-bath correlations are perturbatively negligible compared to the local system energy.

At intermediate or strong coupling, these assumptions fail. The logic gate does not merely exchange energy with an external reservoir; a non-negligible fraction of the total energy resides permanently in the interaction term and the entangled system-bath boundary. Consequently, thermodynamic quantities inferred solely from the reduced density matrix $\rho_S(t)$ become ambiguous.

This subsubsection clarifies precisely why the weak-coupling thermodynamic definitions fail and isolates the specific observables that remain operationally meaningful for evaluating quantum logic gates. This separation ensures that the proposed numerical phase-diagram study yields mathematically defensible and physically interpretable results.

1. The First Law at the Global Level. The rigorous starting point is the exact Hamiltonian decomposition:

$$H_{tot}(t) = H_S(t) \otimes \mathbb{I}_B + \mathbb{I}_S \otimes H_B + H_{int}. \tag{33}$$

For autonomous logic gates, the bare system Hamiltonian H_S is time-independent ($dH_S/dt = 0$). The global density matrix $\rho_{tot}(t)$ evolves unitarily, and the total energy is strictly conserved:

$$\frac{d}{dt}\mathrm{Tr}(\rho_{tot}(t)H_{tot}) = 0. \tag{34}$$

We can formally partition the total energy into three expectation values:

$$E_{tot} = E_S^{bare}(t) + E_B(t) + E_{int}(t), \tag{35}$$

where $E_S^{bare}(t) = \mathrm{Tr}_S(\rho_S(t)H_S)$, $E_B(t) = \mathrm{Tr}(\rho_{tot}(t)H_B)$, and $E_{int}(t) = \mathrm{Tr}(\rho_{tot}(t)H_{int})$. At strong coupling, the exact first-law balance dictates:

$$\frac{dE_S^{bare}}{dt} + \frac{dE_B}{dt} + \frac{dE_{int}}{dt} = 0. \tag{36}$$

Equation (36) reveals the fundamental obstruction: the energy lost by the thermal bath ($-dE_B/dt$) is not equal to the energy gained by the bare logic gate (dE_S^{bare}/dt). A dynamic, non-negligible energy flux continuously flows into and out of the interaction sector $E_{int}(t)$.

2. The Breakdown of the Reduced Heat Current. In the weak-coupling GKSL framework, the heat current from the bath is universally defined as $J_Q^{wc} = \mathrm{Tr}(H_S\dot{\rho}_S)$. At strong coupling, replacing $\dot{\rho}_S$ with the exact non-Markovian derivative from the Nakajima-Zwanzig equation does *not* yield the true thermodynamic heat current.

First, $\mathrm{Tr}(H_S\dot{\rho}_S)$ tracks only the change in the bare system energy. It entirely misses the energy required to establish the system-bath correlations necessary for the gate's operation.

Second, even at thermal equilibrium with a single bath at inverse temperature β, the reduced state of the logic gate is not the canonical Gibbs state of H_S. Due to the non-perturbative interaction, the exact reduced equilibrium state is:

$$\rho_S^{eq} = \mathrm{Tr}_B\left(\frac{e^{-\beta(H_S+H_B+H_{int})}}{Z_{tot}}\right) \neq \frac{e^{-\beta H_S}}{\mathrm{Tr}_S(e^{-\beta H_S})}. \tag{37}$$

Because the steady state deviates from the bare Gibbs state, assigning thermodynamic fluxes based on H_S alone leads to apparent violations of the Second Law (e.g., non-zero continuous heat currents even in single-bath equilibrium).

3. The Hamiltonian of Mean Force. To systematically account for the energetic shift induced by the environment, strong-coupling thermodynamics introduces the **Hamiltonian of Mean Force** (HMF), denoted $H_S^*(\beta)$. It is defined such that the exact reduced equilibrium state retains a generalized canonical form:

$$\rho_S^{eq} = \frac{e^{-\beta H_S^*(\beta)}}{\mathrm{Tr}_S(e^{-\beta H_S^*(\beta)})}. \tag{38}$$

Explicitly, tracing out the bath yields:

$$H_S^*(\beta) = -\frac{1}{\beta}\ln\left[\frac{\mathrm{Tr}_B(e^{-\beta(H_S+H_B+H_{int})})}{\mathrm{Tr}_B(e^{-\beta H_B})}\right]. \tag{39}$$

The HMF encapsulates the free-energetic effect of the bath on the subsystem. Crucially, $H_S^*(\beta)$ is strictly temperature-dependent. It is an effective thermodynamic potential, not a fundamental mechanical observable.

While the HMF elegantly resolves equilibrium strong-coupling thermodynamics, it fails for the autonomous thermodynamic logic gates defined in Section 1.1.2. The NOT gate operates by coupling simultaneously to *multiple* baths at different temperatures (T_C and T_H). In such a non-equilibrium steady state (NESS), there is no single global temperature β, and the HMF cannot be uniquely defined. Consequently, defining a renormalized "system energy" for the actively computing logic gate is formally ambiguous.

4. Operationally Safe Observables for Thermodynamic Computing. Given that standard thermodynamic accounting breaks down, how can we rigorously evaluate the performance of a non-Markovian logic gate? The solution is to strictly separate the logical observables (which depend only on ρ_S) from the energetic observables (which require knowledge of ρ_{tot}).

For the proposed tensor-network research program, we restrict our primary analysis to observables that remain exact and physically unambiguous regardless of the coupling strength:

1. **Logic Fidelity and Output Populations:** The steady-state population of the output qubit $p_{out} = \mathrm{Tr}(\rho_{Y,ss} |1\rangle \langle 1|)$ and the trace distance $\mathcal{F}_{logic} = \frac{1}{2}\left\|\rho_{Y|0} - \rho_{Y|1}\right\|_1$ are defined entirely by the exact reduced density matrix. They remain the definitive metric of computational success.

2. **Gate Switching Time:** The relaxation timescale τ_{relax} required for the reduced density matrix to converge to its non-equilibrium steady state following an input change.

3. **Exact Bath Heat Currents (Optional but Rigorous):** If the tensor network algorithm (e.g., PT-MPO) tracks the bath degrees of freedom explicitly, the true thermodynamic heat current extracted from bath α can be computed rigorously as the rate of change of the bare bath energy:
$$J_{Q,\alpha}^{exact}(t) = -\frac{d}{dt}\mathrm{Tr}\big(\rho_{tot}(t) H_{B_\alpha}\big). \tag{40}$$

5. Summary and Methodological Policy. The breakdown of weak-coupling thermodynamics imposes a strict methodological constraint on the proposed research. We will *not* attempt to compute the "entropy production" or "dissipated heat" of the non-Markovian logic gate using the Lindbladian formulas of Section 1.1.2. Such calculations would be physically meaningless.

Instead, the core scientific question of the project is strictly operational: **As the bath correlation time τ_B and coupling strength α increase, does the trace distance $\mathcal{F}_{logic}$ of the autonomous NOT gate collapse?** By focusing entirely on the logical distinguishability of the exact reduced steady state, the research program avoids the ambiguities of strong-coupling thermodynamics while directly addressing the computational viability of thermodynamic logic in realistic environments.

To execute this exact calculation, we must evaluate the non-Markovian influence functional without truncation. Section 3 introduces the Process Tensor formalism, which systematically maps this formidable path-integral problem into an efficient tensor-network contraction.

3. Tensor Network Formulations for Open Quantum Systems

The conceptual framework of Section 2 establishes that non-Markovian memory invalidates the Lindblad picture, and that the operational consequences are fully captured by the exact process tensor. However, writing down the exact influence functional or process tensor is merely a formal solution. For structured environments with finite correlation times τ_B, the memory kernel does not truncate, leading to an exponential explosion in the number of time histories that must be integrated.

This section introduces the numerical methodologies required to compute the exact process tensor efficiently. We translate the temporal correlations into a one-dimensional many-body problem along the time axis, which can then be systematically compressed using Matrix Product Operators (MPO).

3.1. From Path Integrals to Process Tensors

3.1.1 The Process Tensor Formalism and Multi-Time Correlations

Section 2 established that when the bath correlation time τ_B is finite, the reduced dynamics of the logical subsystem becomes history-dependent. However, characterizing this non-Markovian memory solely through a one-parameter dynamical map $\Lambda_{t:0}$—which merely evolves an initial state $\rho_S(0)$ to a final state $\rho_S(t)$—is operationally insufficient.

In a computational setting, an experimenter (or an automated protocol) actively intervenes on the system. Logic inputs are switched, error-correction pulses are applied, and mid-circuit measurements are performed. When an intervention alters the system state at time t_k, it simultaneously alters the subsequent back-action from the environment, because the environment is entangled with the pre-intervention system state. A standard dynamical map erases this crucial system-environment correlation. To rigorously evaluate a thermodynamic logic gate under arbitrary input switching, we require a formalism that predicts the exact response of the open system to arbitrary sequences of control operations.

The **Process Tensor** provides exactly this mathematical object. It is the multi-time generalization of a quantum channel. It maps an entire sequence of interventions to the final density matrix, encapsulating all temporal correlations and environmental memory in a rigorously causal framework.

1. The Discrete-Time Operational Setup. We discretize the total computational time into n steps: $t_0 < t_1 < \cdots < t_n$, separated by Δt. The joint system-environment state initially factorizes as $\rho_{SE}(t_0) = \rho_S(t_0) \otimes \rho_E^{eq}$.

Between any two consecutive times t_j and t_{j+1}, the global system evolves unitarily according to the joint propagator:

$$U_{j+1:j} = \exp\left(-iH_{tot}\Delta t\right). \tag{41}$$

At each discrete time step t_j, an experimenter may apply an instantaneous intervention, represented mathematically as a Completely Positive Trace-Preserving (CPTP) map $\mathcal{A}_j$ acting exclusively on the system Hilbert space $\mathcal{B}(\mathcal{H}_S)$. These interventions can represent unitary clock pulses, projective readouts, or the switching of the logical input baths.

The exact reduced density matrix of the logic gate at the final time t_n, conditional on the specific sequence of chosen interventions $\mathbf{A}_{n-1:0} = (\mathcal{A}_{n-1}, \ldots, \mathcal{A}_0)$, is given by:

$$\rho_S(t_n|\mathbf{A}_{n-1:0}) = \mathrm{Tr}_E\left[U_{n:n-1}\mathcal{A}_{n-1}U_{n-1:n-2}\ldots\mathcal{A}_0(\rho_S(t_0)\otimes\rho_E^{eq})\right]. \tag{42}$$

Notice that the system state at t_n is a multilinear function of the applied maps $\mathcal{A}_j$.

2. Definition and Choi Representation of the Process Tensor. Because Eq. (42) is strictly multilinear, we can abstract the environmental dynamics away from the specific choice of system operations. We define the Process Tensor $\mathcal{T}_{n:0}$ as the multilinear superchannel that accepts the sequence of operations and outputs the final state:

$$\rho_S(t_n|\mathbf{A}_{n-1:0}) = \mathcal{T}_{n:0}[\mathcal{A}_{n-1}, \dots, \mathcal{A}_0]. \tag{43}$$

By invoking the Choi-Jamiołkowski isomorphism, any CPTP map $\mathcal{A}_j$ can be represented as a positive semidefinite matrix A_j acting on a doubled Hilbert space $\mathcal{H}_S \otimes \mathcal{H}_S$. Consequently, the entire Process Tensor $\mathcal{T}_{n:0}$ can be uniquely represented as a many-body positive operator $\Upsilon_{n:0}$ living on the tensor product of all time slots.

The final system state is computed via a generalized Born rule, contracting the Process Tensor with the chosen interventions:

$$\rho_S(t_n|\mathbf{A}_{n-1:0}) = \mathrm{Tr}_{0\dots n-1}\left[\Upsilon_{n:0}\left(\mathbb{I}_n \otimes A_{n-1}^T \otimes \dots \otimes A_0^T\right)\right], \tag{44}$$

where the partial trace is taken over all past intervention slots, leaving an uncontracted index corresponding to the final output state at t_n.

3. Causality and the Definition of Markovianity. To represent a physical environment, the many-body operator $\Upsilon_{n:0}$ must satisfy strict causality constraints: future interventions cannot influence past statistics. Mathematically, this enforces a nested hierarchy of partial traces: tracing out the final output slot deterministically yields the Process Tensor for the sequence up to t_{n-1}, independent of any choice of $\mathcal{A}_{n-1}$.

Within this exact operational framework, the definition of a Markovian environment becomes mathematically precise. An environment is completely Markovian if and only if the Process Tensor factorizes perfectly across time slots:

$$\Upsilon_{n:0}^{Markov} = \Lambda_{n:n-1} \otimes \Lambda_{n-1:n-2} \otimes \dots \otimes \Lambda_{1:0} \otimes \rho_S(t_0), \tag{45}$$

where $\Lambda_{j:j-1}$ are independent completely positive maps. In this limit, an intervention $\mathcal{A}_k$ at time t_k modifies the state, but all subsequent evolution $\Lambda_{j>k}$ is entirely oblivious to what the state was *before* the intervention. The bath has no memory.

Non-Markovianity is thus rigorously defined as the non-separability (temporal entanglement) of the Choi state $\Upsilon_{n:0}$. The environment generates correlations across distinct time steps.

4. Multi-Time Observables and Thermodynamic Logic. The ability to compute multi-time correlations is not a formal luxury; it is the central requirement of the proposed research. Consider testing the logic fidelity of the thermodynamic NOT gate. The standard protocol requires switching the input temperature T_{in} from Cold to Hot at t_0, and measuring the output state at a later time t_m.

However, if we wish to evaluate the *dynamic robustness* of the gate, we might rapidly switch the input from Cold $\to$ Hot $\to$ Cold at times (t_0, t_k). The output at t_m depends non-trivially on the finite memory time τ_B. If $\tau_B > t_k - t_0$, the bath "remembers" the initial Cold state and interferes with the thermalization toward the new Hot state, potentially causing the logic gate to undergo hysteresis or coherent errors.

Using the Process Tensor, computing the result of this rapid switching protocol requires zero additional simulation of the bath. Once $\Upsilon_{n:0}$ is computed for the specific structured environment, the experimenter simply contracts it with different sequences of input operations $\{A_j^T\}$ via Eq. (44) to map the entire operational phase diagram of the logic gate.

5. Mapping the Influence Functional to a Tensor Network. While the Process Tensor isolates the environmental memory perfectly, computing the many-body operator $\Upsilon_{n:0}$ via exact diagonalization of the global Hamiltonian H_{tot} is impossible due to the infinite degrees of freedom in the bath.

The breakthrough comes from recognizing that the Process Tensor $\Upsilon_{n:0}$ is the exact discrete-time equivalent of the Feynman-Vernon influence functional $\mathcal{F}[x, x']$ derived in Section 2.1.1. In the path integral formalism, the bath induces long-range pairwise interactions in time, weighted by the bath correlation function $C(t_i - t_j)$.

By organizing these temporal correlations sequentially, the n-slot Process Tensor can be cast into the structure of a one-dimensional **Matrix Product Operator (MPO)** lying along the time axis. The "bond dimension" χ of this tensor network quantifies the amount of quantum memory the bath carries from the past to the future.

Section 3.1.2 will detail this exact numerical compression scheme (PT-MPO), explaining how the exponential explosion of memory is tamed algorithmically, allowing us to exactingly simulate the thermodynamic logic gate in the strong-coupling, non-Markovian regime.

3.1.2 PT-MPO Compression, OQuPy Implementation, and Bond Dimension Scaling

The formal equivalence between the discrete-time Feynman-Vernon influence functional and the many-body process tensor $\Upsilon_{n:0}$ identifies the exact nature of the computational bottleneck: the environment induces temporal correlations between the logic gate's state at time t_i and its state at all other times t_j. A naive evaluation of these correlations over n time steps scales exponentially as $\mathcal{O}(d^{2n})$, where d is the local Hilbert space dimension of the system.

This subsubsection outlines the numerical algorithm designed to break this exponential scaling: the Time-Evolving Matrix Product Operator (TEMPO) method and its generalization to the Process Tensor (PT-MPO). We define the tensor network compression of the memory kernel, analyze the computational complexity controlled by the bond dimension χ, and specify the parameters required for executing the research program using the open-source library `OQuPy`.

1. The Matrix Product Operator (MPO) Structure in Time. The process tensor $\Upsilon_{n:0}$ is a positive operator acting on the $2n$-fold tensor product space $\bigotimes_{k=0}^{n-1}(\mathcal{H}_{S_k} \otimes \mathcal{H}_{S_k}^*)$. We map this temporal structure to a one-dimensional spatial lattice, where each "site" corresponds to a discrete time step t_k.

In this mapping, the exact process tensor can be written as a Matrix Product Operator:

$$\Upsilon_{n:0} = \sum_{\{\alpha_k, \beta_k\}} \mathrm{Tr}\Big(B^{[n-1]}_{\alpha_{n-1},\beta_{n-1}} \dots B^{[1]}_{\alpha_1,\beta_1} B^{[0]}_{\alpha_0,\beta_0}\Big) \, |\alpha_{n-1} \dots \alpha_0\rangle\langle\beta_{n-1} \dots \beta_0|, \tag{46}$$

where $\{\alpha_k, \beta_k\}$ are the physical indices corresponding to the Liouville-space basis states of the system at time t_k (taking values from 1 to d^2), and $B^{[k]}_{\alpha_k,\beta_k}$ are $\chi_k \times \chi_{k-1}$ complex matrices.

The internal matrix dimensions, bounded by the **bond dimension** $\chi = \max_k\{\chi_k\}$, dictate the rank of the tensor network. Physically, the bond dimension quantifies the size of the effective quantum memory required to transmit the bath's influence from the past ($t < t_k$) to the future ($t > t_k$). If the bath is perfectly Markovian, it carries zero memory across time steps, and $\chi = 1$. The non-Markovian history dependence manifests as $\chi > 1$.

2. The Influence Tensor and Memory Truncation. For an environment composed of independent harmonic oscillators linearly coupled to the system, the influence functional is exactly solvable. Using a symmetric Trotter splitting of the joint unitary propagator $U(t_{k+1}, t_k) = \exp(-iH_S\Delta t/2)\exp(-i(H_B + H_{int})\Delta t)\exp(-iH_S\Delta t/2) + \mathcal{O}(\Delta t^3)$, the temporal correlations decouple into a product of pairwise influence tensors $I_{k,m}$. The tensor $I_{k,m}$ couples the system state at time t_m to the state at a later time t_k, weighted by the discrete integral of the bath correlation function $C(t_k - t_m)$.

The exact PT-MPO is constructed by sequentially multiplying these pairwise influence tensors into the network. Because $C(t)$ decays over the bath correlation time τ_B, the influence between widely separated times becomes small. We introduce a temporal truncation parameter, the **memory length** K, such that:

$$I_{k,m} \approx \mathbb{I} \quad \text{for} \quad k - m > K. \tag{47}$$

The integer K is chosen such that $K\Delta t \gg \tau_B$. Under this cutoff, the environment retains full non-Markovian memory up to the time $\tau_{mem} = K\Delta t$, and acts as a Markovian reservoir beyond it.

3. Algorithmic Compression and SVD Tolerance. Even with a finite memory length K, exact contraction of the network leads to a bond dimension scaling as $\chi \sim d^{2K}$. For realistic correlation times requiring $K \sim 100$, exact evaluation is impossible.

The PT-MPO algorithm relies on the fact that the temporal entanglement generated by physical baths is not maximal. The matrices $B^{[k]}$ contain redundant information. Following standard Matrix Product State algorithms (e.g., TEBD), we sequentially compress the tensor network at each time step using the Singular Value Decomposition (SVD).

At step t_k, the network is bipartite into "past" and "future". The SVD of the boundary tensor yields a diagonal matrix of singular values $\Lambda = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_r)$, sorted in descending order. The square of these singular values represents the probability weight of the temporal correlations. We discard the singular values that fall below a specified tolerance threshold ϵ:

$$\frac{\sum_{i>\chi} \lambda_i^2}{\sum_i \lambda_i^2} \leq \epsilon. \tag{48}$$

By keeping only the χ largest singular values, the bond dimension is strictly bounded. The overall computational complexity of constructing the PT-MPO scales polynomially as $\mathcal{O}(nK\chi^3 d^6)$.

For the proposed research, demonstrating numerical convergence requires showing that the logical observables (e.g., trace distance) are invariant under a simultaneous decrease in Δt, an increase in K, and a decrease in the SVD tolerance ϵ.

4. The OQuPy Implementation Strategy. Constructing the PT-MPO from the raw influence functional requires careful management of Trotter errors and tensor contractions. Rather than coding this infrastructure from scratch, the proposed research program will utilize `OQuPy` (Open Quantum Python), a high-performance library engineered for simulating non-Markovian dynamics using the TEMPO and PT-MPO algorithms.

For the baseline investigations, `OQuPy` natively handles:

- **Structured Spectral Densities:** Generating the influence tensors $I_{k,m}$ from arbitrary inputs of $J(\omega)$ (e.g., Ohmic or sub-Ohmic with exponential cutoffs).
- **SVD Compression:** Managing the tensor network contractions under a specified tolerance ϵ, outputting the compressed process tensor $\Upsilon_{n:0}$ as a manipulable Python object.
- **Generalized Born Contraction:** Allowing the user to input arbitrary sequences of logical control maps $\mathbf{A}_{n-1:0}$ and returning the final exact reduced density matrix.

5. Scaling the Thermodynamic Logic Gate to Multiple Baths. The fundamental challenge of simulating the complete autonomous NOT gate described in Section 1.1.2 is its composite structure. The 3-qubit machine (X, Y, W) interacts simultaneously with three independent thermal environments $(Bath_X, Bath_Y, Bath_W)$.

Because the three environments are statistically independent, their exact path-integral influence functionals factorize:

$$\mathcal{F}_{total} = \mathcal{F}_X \times \mathcal{F}_Y \times \mathcal{F}_W. \tag{49}$$

Algorithmically, the global process tensor $\Upsilon_{n:0}^{global}$ can be constructed by independently generating the PT-MPO for each structured bath using `OQuPy`, and then taking their tensor product.

However, the internal dimension of the system is $d = 2^3 = 8$. The local Liouville-space dimension per time step is $d^2 = 64$. When the three independent PT-MPOs are contracted with the full 3-qubit Hamiltonian dynamics, the effective bond dimension χ_{global} will grow rapidly due to the entanglement generated between the qubits by the internal interaction V_{int}.

Caveat 2 (The Bond Dimension Explosion)**.** *The primary technical risk of the proposed research lies in simulating the full 3-qubit gate coupled to three non-Markovian baths. Even with SVD compression, the required bond dimension χ_{global} may exceed available computational memory limits before reaching the non-equilibrium steady state at $t \to \infty$. To mitigate this, Section 4 specifies a rigid, phased execution plan. The research must begin by identifying the non-Markovian failure threshold of a single logical qubit coupled to a single structured bath. Only after this Minimal Publishable Unit is secured will we attempt the full multi-bath tensor contraction.*

4. Proposed Research Program and Execution

The preceding sections have established the physical breakdown of the Markovian assumption for logic gates and formulated the PT-MPO tensor network machinery required to simulate the exact dynamics. We now synthesize these theoretical developments into a concrete research program designed for a Master's project. The scope is calibrated to an 8-course equivalent workload (approximately 320 hours).

The program is structured hierarchically. It begins with the replication of the baseline Markovian logic gate, proceeds to the quantification of single-qubit memory effects, and concludes with the phase-diagram mapping of the autonomous, multi-qubit NOT gate. This phased approach ensures a structured progression and aims at producing results suitable for journals.

4.1. Stress-Testing the Thermodynamic NOT Gate

The primary objective is to address an open question in the thermodynamics of computation: *As the bath correlation time τ_B increases beyond the system relaxation time τ_R, at what critical* *coupling strength does the logical distinguishability (trace distance) of an autonomous thermodynamic NOT gate fall below the classical error-correction threshold?*

4.1.1 Analytical Setup of the NOT Gate under Structured Baths and Phase Diagram Mapping

The research program is partitioned into three sequential projects. Each project corresponds to roughly one-third of the total workload, blending analytical derivation with numerical simulation using the `OQuPy` library.

Project 1: The Markovian Baseline and Global GKSL Simulation. The objective of the first phase is to independently verify the weak-coupling analytical results of Section 1.1.2 using standard master-equation solvers.

- **Analytical Work:** Derive the global jump operators for the 3-qubit interaction Hamiltonian $V_{int} = g(\sigma_-^{(X)}\sigma_-^{(Y)}\sigma_+^{(W)} + h.c.)$. Construct the dissipators $\mathcal{D}_k$ without making the local secular approximation, which is known to violate thermodynamic consistency.
- **Numerical Work:** Integrate the global GKSL master equation to the non-equilibrium steady state (NESS). Calculate the steady-state output population p_Y as a function of the input temperature $T_X \in [T_C, T_H]$.
- **Deliverable:** Compute the logic fidelity $\mathcal{F}_{logic}(T_X, g, \gamma)$ curve. Identify the weak-coupling ratio g/γ that maximizes the trace distance before the onset of the quantum Zeno effect, where strong internal coupling g suppresses thermalization.

Project 2: Single-Qubit Memory Effects and the PT-MPO Workflow. Before attempting the composite 3-qubit gate, the student will implement the PT-MPO formalism on a single qubit coupled to a non-Markovian bath to establish methodological control.

- **Analytical Work:** Define Ohmic and sub-Ohmic spectral densities $J(\omega)$ with an exponential cutoff ω_c. Calculate the exact bath correlation time τ_B as a function of the cutoff frequency.
- **Numerical Work:** Use `OQuPy` to construct the PT-MPO $\Upsilon_{n:0}$ for a single qubit undergoing driven evolution. Perform a convergence analysis: systematically reduce the time step Δt, increase the memory length K, and decrease the SVD tolerance ϵ until the final density matrix $\rho_S(t_n)$ converges within a specified tolerance.
- **Deliverable:** A quantitative map of the single-qubit relaxation rate versus the coupling strength α. The student will computationally identify the onset of non-Markovian revivals (coherent oscillations preceding thermalization) when $\alpha \sim \omega_c$. This establishes the computational boundaries (maximum tractable bond dimension χ) of the available computing resources.

Project 3: Phase Diagram of the Non-Markovian NOT Gate. This is the core research objective. The student will combine the 3-qubit Hamiltonian of Project 1 with the structured bath PT-MPOs of Project 2.

- **Analytical Work:** Formulate the multi-bath process tensor. Because the baths X, Y, W are statistically independent, their influence functionals factorize. The global non-Markovian

NESS is obtained by contracting the three independent PT-MPOs with the joint 3-qubit unitary operations.

- **Numerical Work:** Fix the temperatures $T_W = T_H$ and sweep $T_X \in \{T_C, T_H\}$. For a fixed Ohmic spectral density, calculate the exact reduced density matrix of the output qubit Y in the long-time limit.
- **Deliverable (Phase Diagram):** Generate a 2D contour plot mapping the logic fidelity $\mathcal{F}_{logic}$ against the system-bath coupling strength α and the non-Markovianity parameter τ_B/τ_R. The phase boundary delineates the regime where environmental memory degrades the thermodynamic logic encoding.

4.1.2 Caveats, Alternative Geometries (TTN), and Mitigation Strategies

The primary bottleneck in Project 3 is the growth of the internal bond dimension χ_{global} when contracting three independent PT-MPOs across a 3-qubit Hilbert space. If the required bond dimension exceeds available memory, the truncation tolerance ϵ will artificially enforce Markovianity, rendering the results physically inaccurate.

To ensure the project remains viable and yields rigorous results, the following mitigation strategies are incorporated into the research plan:

1. Pivot to Single-Qubit Fluctuation Theorems. If the 3-qubit, 3-bath simulation fails to converge, the student will pivot to a mathematically rigorous 1-qubit thermodynamic protocol. By periodically driving a single qubit with a classical control field, the student can evaluate the exact non-Markovian entropy production versus switching speed. This maps to Jarzynski-type fluctuation theorems and provides a publishable result on the energy-speed tradeoff in strongly coupled open systems, circumventing the multi-bath computational limit.

2. Tree Tensor Networks (TTN) for Multi-Bath Environments. If the standard 1D sequential contraction in `OQuPy` encounters a bond dimension threshold, the student will investigate restructuring the tensor contraction into a Tree Tensor Network (TTN) geometry. Instead of a single chronological chain, the temporal correlations from the three baths can be contracted hierarchically. While implementing a bespoke TTN solver may exceed the scope of a 1-year project, analyzing the theoretical memory scaling χ_{TTN} versus χ_{MPO} for this specific 3-bath setup constitutes a robust mathematical physics analysis suitable for the thesis.

3. Reaction-Coordinate Benchmarking. To validate the exact PT-MPO results at strong coupling, the student will implement a Reaction Coordinate (RC) mapping. By absorbing the primary collective mode of the Ohmic bath into the system Hamiltonian, the student will create an effective 4-qubit system coupled to a weakly interacting, Markovian residual bath. Simulating this enlarged system via a standard GKSL solver (as in Project 1) provides an independent analytical bound against which the PT-MPO phase diagram (Project 3) can be checked in the strong-coupling regime.

5. Project Feasibility, Novelty, and Risk Assessment

This section provides a rigorous evaluation of the proposed research program, intended specifically for internal review among the supervisory team. It assesses the novelty of the proposal against the current literature, defines the precise risk factors, and maps the computational and theoretical requirements to the specific expertise of the collaborators.

5.1. Literature Audit and Novelty Assessment

The proposed research synthesizes three mature but currently disjoint subfields:

1. **Autonomous Quantum Thermal Machines:** Recent formalisms establish the architecture for thermodynamic computing (e.g., autonomous Quantum Processing Units and thermodynamic neurons). However, these architectures rely entirely on the weak-coupling, Markovian Lindblad limit.

2. **Non-Markovian Tensor Networks:** The Process Tensor Matrix Product Operator (PT-MPO) framework has reached maturity for simulating structured environments, with robust open-source implementations now available (e.g., `OQuPy`).

3. **Strong-Coupling Thermodynamics:** The conceptual framework for renormalizing observables (Hamiltonian of mean force, exact bath energy accounting) is established but has not been applied to actively switching logic gates.

The novelty of this proposal lies strictly in their intersection. Current literature confirms that no existing work evaluates the computational fidelity of thermodynamic logic under structured, non-Markovian bath dynamics via tensor networks. The high barrier to entry—requiring simultaneous expertise in quantum thermodynamics, open-system numerics, and computational logic—results in a low risk of competing groups publishing identical synthesis work (scoop risk).

5.2. Team Synergy and Task Allocation

The project's feasibility depends on flexible division of labor aligned with leveraging our strengths:

- **Theoretical Physics and Thermodynamics:** Responsible for enforcing thermodynamic consistency at strong coupling. This includes defining the operationally safe observables (trace distance, logic fidelity) and preventing the misapplication of weak-coupling heat and entropy formulas to the PT-MPO data.

- **Tensor Network Architectures:** Provides oversight for the tensor network contraction strategies. If the standard 1D PT-MPO bond dimension exceeds computational limits during the multi-bath simulation, Dr. Mukherjee's expertise in Tree Tensor Networks (TTN) and hierarchical equations of motion will guide the structural mitigation strategies.

- **Numerical Execution:** Leverages coding and machine learning background to implement the `OQuPy` simulations, execute the convergence testing, and generate the phase diagrams. By utilizing established libraries, the student is insulated from fundamental algorithmic development and can focus entirely on physical application and data generation.

5.3. Detailed Risk and Mitigation Matrix

The following table summarizes the primary risk vectors and their corresponding mitigation strategies, ensuring a reasonable output is achieved regardless of computational bottlenecks.

Risk Vector	**Caveat**	**Mitigation Strategy**
Bond-Dimension Explosion	The 3-qubit, 3-bath contraction generates temporal entanglement that exceeds available computational memory.	Pivot to TTN geometries or restrict the final deliverable to the 1-qubit exact entropy production model.
Thermodynamic Ambiguity	Evaluating ill-defined "heat" or "dissipation" at strong coupling leads to unphysical or incorrect claims.	Restrict primary claims to strictly operational logic observables (reduced-state trace distance, gate switching times).
Absence of Sharp Phase Boundary	The transition from Markovian success to non-Markovian failure may manifest as a smooth crossover.	Frame the objective as a continuous characterization of logical robustness rather than a search for a critical phase transition point.

Table 1: Summary of project risks, caveats, and structural mitigations for the Master's project.

6. Reading List

This reading list is a deliberately selective project-start note. It does not attempt to survey all of quantum thermodynamics, open quantum systems, or tensor networks. Its purpose is narrower: to identify the smallest serious body of literature needed to begin a project on thermodynamic quantum logic gates, especially in the direction of structured baths, non-Markovian dynamics, and tensor-network or process-tensor numerics.Three principles shape the reading list.

1. The baseline model must be learned first. The central starting point is the thermodynamic-neuron paper of Lipka-Bartosik, Perarnau-Llobet, and Brunner, which formulates logic gates such as NOT, NOR, and 3-MAJORITY using autonomous quantum thermal machines. Any extension should be built relative to that model, not in the abstract.

2. ArXiv versions are preferred whenever they were directly verified and carry the same scientific content. This is especially useful when appendices, notation, or technical exposition are more transparent there.

3. Reading must be tied to action. Every block below is paired with a concrete technical outcome: a baseline derivation, a reproduced figure, a benchmark code, or a thermodynamic-audit note.

6.1. What is worth reading first

Level	Source	Why it is here	What must come out of it
A	Potts, *Introduction to Quantum Thermodynamics*	Cleanest short entry into weak-coupling quantum thermodynamics	Comfort with heat, work, entropy production, CPTP maps, and Markovian master equations
A	Tavakoli, *Quantum information in quantum thermal machines*	Gives the correct autonomous-thermal-machine mindset	Understanding of thermal machines as information-processing devices
B	Lipka-Bartosik–Perarnau-Llobet–Brunner, *Thermodynamic Computing via Autonomous Quantum Thermal Machines*	The actual baseline gate model	A fully reconstructed NOT/NOR gate model and a list of approximations
C	Fux, *Process tensor networks for non-Markovian open quantum systems*	Best extended entry into PT-MPO and memory effects	A precise picture of what a process tensor is and why it is better than ad hoc memory corrections
C	OQuPy package paper	Practical entry point for non-Markovian numerics	Decision on whether the first beyond-Markovian benchmark can be done with existing software
D	Rivas, *Strong Coupling Thermodynamics of Open Quantum Systems*	Prevents thermodynamic overclaiming beyond Lindblad	A list of quantities that remain safe to interpret beyond weak coupling

6.2. Annotated reading list

6.2.1 Level A: orientation texts that should be read immediately

1. Patrick P. Potts, *Introduction to Quantum Thermodynamics*, arXiv:1906.07439
Difficulty: Introductory **Urgency: Essential**

Why read it. This is the most efficient entry point into the thermodynamic language that is actually used in the gate literature. It covers reduced dynamics, entropy production, heat and work in open quantum systems, and the weak-coupling Markovian setting without requiring a full textbook investment.

Read it for.

- The operational meaning of heat, work, and entropy production in reduced-system descriptions.
- The role of Markovianity and complete positivity.
- The difference between externally driven setups and autonomous machines.

Do not get lost in. Broad philosophical discussion. The practical goal is simply to become fluent enough to parse the thermodynamic-computing literature.

Concrete output after reading. A one-page note fixing the notation for energy, currents, and reduced dynamics to be used in the project.

2. Armin Tavakoli, *Lecture notes: Quantum information in quantum thermal machines*
Difficulty: Introductory/ Intermediate **Urgency: Essential**

Why read it. This is the right conceptual bridge from quantum thermodynamics to autonomous thermal devices. It gives the correct operational viewpoint for reading later papers on thermodynamic neurons and thermal logic.

Read it for.

- The architecture of autonomous thermal machines.
- Resource-like reasoning with temperature biases, reservoirs, and information tasks.
- The language used in thermal-machine papers that often appears without much pedagogical warning.

Do not get lost in. Peripheral quantum-information topics that do not feed directly into the gate model.

Concrete output after reading. A short summary of what makes an autonomous thermal machine different from a driven quantum control protocol.

3. Camille Latune, *Introduction to Quantum Thermodynamics* (slides) Difficulty: Introductory **Urgency: Optional**

Why read it. This is not a core source, but it is useful as a compact map of the field. It can be skimmed in one sitting and helps place the main project in a larger context.

Read it for.

- Vocabulary.
- A taxonomy of subfields.
- Orientation, not technical depth.

6.2.2 Level B: the baseline gate literature

4. P. Lipka-Bartosik, M. Perarnau-Llobet, and N. Brunner, *Thermodynamic Computing via Autonomous Quantum Thermal Machines*, arXiv:2308.15905
Difficulty: Intermediate **Urgency: Essential**

Why read it. This is the baseline paper for the entire project. It introduces thermodynamic neurons built from few-qubit autonomous thermal machines, and constructs logic gates such as NOT, NOR, and 3-MAJORITY. It is the model that must first be reproduced before any structured-bath or non-Markovian extension is attempted.

Read it for.

- The exact Hamiltonian and coupling structure.
- How logical inputs are encoded through bath temperatures.
- How outputs are extracted from the thermal machine.
- Which approximations are essential: local master equations, reset models, weak coupling, and steady-state reasoning.

What must be written down explicitly.

- A complete list of assumptions.

- A derivation note for the NOT gate at minimum.
- A list of observables that are operationally defined already in the baseline paper.

Comment. This should be treated as the canonical reference.

6.2.3 Level C: non-Markovian dynamics, process tensors, and tensor methods

5. Regina Finsterhölzl et al., *Using Matrix-Product States for Open Quantum Many-Body Systems: Efficient Algorithms for Markovian and Non-Markovian Time-Evolution*

Difficulty: Intermediate **Urgency: Essential soon**

Why read it. This is the cleanest bridge from standard tensor-network familiarity to open-system tensor numerics. It explains how MPS ideas enter dissipative and non-Markovian dynamics and why direct density-matrix evolution rapidly becomes impractical.

Read it for.

- How Markovian and non-Markovian open-system dynamics are represented in tensor language.
- The logic of chain mappings and reservoir representations.
- Where bond-dimension growth becomes the real obstacle.

Caution. This is not yet the process-tensor formalism, so it should be read as a tensor-method bridge, not as the final formalism for the project.

6. Gerald E. Fux, *Process tensor networks for non-Markovian open quantum systems* (PhD thesis, St Andrews, 2022)

Difficulty: Intermediate/ Advanced **Urgency: Essential**

Why read it. This is arguably the single most useful extended source for the intended non-Markovian direction. It develops the process tensor carefully, explains why ordinary dynamical maps are insufficient in the presence of memory, and shows how PT-MPO compresses the environmental influence into a workable tensor network.

Read it for.

- The process tensor as a multi-time object.
- The construction and use of PT-MPO.
- Time-translation-invariant process tensors and steady-state computation.
- The relation between numerical efficiency and finite memory length.

How to read it. Read the conceptual chapters first, then only the numerically relevant sections. Do not attempt to digest the entire thesis in one pass.

7. *Process Tensor Approaches to Non-Markovian Quantum Dynamics* (perspective/review)

Difficulty: Intermediate **Urgency: Essential soon**

Why read it. This is the concise overview that should be read after the thesis introduction. It helps organize the bigger picture and prevents the formalism from becoming an opaque algorithmic gadget.

Read it for.

- Why the process tensor unifies many non-Markovian approaches.
- The relation between process tensors, tensor-network compression, and multitime observables.
- Where present numerical frontiers actually lie.

8. G. E. Fux et al., *Tensor network simulation of chains of non-Markovian open quantum systems*, arXiv:2201.05529

Difficulty: Advanced **Urgency: Essential soon**

Why read it. This is closer to the kind of architecture the project may eventually need than generic process-tensor reviews. It demonstrates how environmental memory can be compressed before being embedded into a larger coupled problem.

Read it for.

- The scaling logic when one passes from a single open system to coupled subsystems.
- The actual numerical bottlenecks, rather than imagined ones.
- Whether a multi-qubit gate under structured baths is remotely feasible with the chosen method.

9. Gerald E. Fux et al., *OQuPy: A Python package to efficiently simulate non–Markovian open quantum systems with process tensors*, arXiv:2406.16650

Difficulty: Intermediate **Urgency: Essential**

Why read it. This is the practical gateway into the non-Markovian numerics. It explains what OQuPy can do, what assumptions it makes, and where the existing codebase may already cover the first proof-of-principle calculations.

Read it for.

- Supported bath models and spectral densities.
- Single-bath versus multi-bath capabilities.
- Which observables are easy to compute and which are not.
- Whether the first beyond-Markovian gate benchmark can be built with minimal custom infrastructure.

10. *MPSDynamics.jl: Tensor network simulations for finite-temperature environments*, arXiv:2406.07052

Difficulty: Intermediate/ Advanced **Urgency: Optional**

Why read it. This is a useful secondary option if a more tensor-native route is preferred. It should not be read before OQuPy unless there is already a strong Julia preference or a clear numerical reason to do so.

Read it for.

- Comparison with the OQuPy route.
- Finite-temperature environment handling.
- Whether its abstractions are better matched to the eventual gate geometry.

6.2.4 Level D: strong-coupling thermodynamics and interpretation limits

11. Á. Rivas, *Strong Coupling Thermodynamics of Open Quantum Systems*, arXiv:1910.01246

Difficulty: Advanced **Urgency: Essential**

Why read it. This is the main conceptual safeguard against writing thermodynamic nonsense after leaving the weak-coupling Lindblad regime. It explains why system-bath factorization fails and why standard reduced-system identifications of heat, work, and entropy production become subtle.

Read it for.

- Which thermodynamic quantities become scheme-dependent at strong coupling.
- Which statements from weak-coupling thermodynamics survive and which do not.
- How to keep the project centered on operationally safe observables.

Practical implication. This paper should be used to decide, early, whether the project will discuss heat currents in a foundational sense or restrict itself to logic fidelity, relaxation times, and other safer output observables.

12. *Quantum Heat Current under Non-perturbative and Non-Markovian System-Reservoir Couplings*, arXiv:1609.08783

Difficulty: Advanced **Urgency: Essential soon**

Why read it. The intended project is explicitly about logic implemented by thermal response. This paper is directly relevant because it shows how naive current definitions can fail when coupling is non-perturbative and memory effects are real.

Read it for.

- Heat-current definitions beyond perturbation theory.
- The role of interaction energy in current accounting.
- How cautious one must be when interpreting reduced-system observables as thermodynamic currents.

13. Wei-Ming Huang and Wei-Min Zhang, *Strong Coupling Quantum Thermodynamics with Renormalized Hamiltonian and Temperature*, arXiv:2010.01828

Difficulty: Advanced **Urgency: Optional**

Why read it. This is not the first strong-coupling paper to read, but it is a useful supplementary viewpoint. It makes explicit how renormalized Hamiltonians and effective temperatures enter exact open-system thermodynamics.

Read it for.

- Renormalized Hamiltonian and temperature notions.
- How exact steady states may still resemble Gibbs forms after renormalization.
- A sense of how far thermodynamic language can be pushed safely.

6.2.5 Level E: adjacent context and optional expansion

14. *Non-Markovian thermal operations boosting the performance of heat engines*, Phys. Rev. E 106, 014114 (2022)

Difficulty: Intermediate/ Advanced **Urgency: Optional**

Why read it. This is not a gate paper, but it shows that memory effects can matter operationally in autonomous thermal devices. It helps calibrate expectations for what kinds of improvements or degradations non-Markovianity may plausibly generate.

15. Miika Rasola and Mikko Möttönen, *Autonomous Quantum Heat Engine Based on Non-Markovian Dynamics of an Optomechanical Hamiltonian*, arXiv:2403.18515
Difficulty: Intermediate/ Advanced **Urgency: Optional**

Why read it. This paper provides a concrete example of a non-Markovian autonomous thermal device. It is useful not because it solves the gate problem, but because it shows how structured environments and autonomous operation can coexist in a nontrivial model.

6.3. Execution plan tied to the reading

6.3.1 Phase I: first two weeks

Read: Potts, Tavakoli, and the baseline gate paper.

Output required:

1. A two- to four-page technical note reconstructing the baseline NOT gate.
2. A list of every approximation used in the thermodynamic-neuron paper.
3. A clear statement of which observable encodes the logical output.

6.3.2 Phase II: weeks 3–4

Read: Finsterhölzl et al. and the OQuPy paper.

Output required:

1. A decision memo on whether the first non-Markovian benchmark can be done in OQuPy.
2. A minimal software prototype for a single open qubit coupled to a structured bath.

6.3.3 Phase III: month 2

Read: Fux thesis introduction and the process-tensor perspective.

Output required:

1. A one-page explanation of the process tensor in plain but technically correct language.
2. A benchmark demonstrating recovery of the Markovian limit as the memory time is decreased.

6.3.4 Phase IV: months 3–4

Read: the chain-of-non-Markovian-open-systems paper and selected sections of the strong-coupling literature.

Output required:

1. A feasibility assessment of whether a multi-qubit gate is numerically realistic.
2. A thermodynamic-audit note listing which observables remain safe to interpret beyond weak coupling.

6.3.5 Phase V: thereafter

From this point onward, reading should become question-driven. New papers should be added only when a calculation raises a specific formal or numerical issue.

6.4. A minimal core set if time is tight

If the project must start quickly, the following six items are enough to begin intelligently:

1. Potts, arXiv:1906.07439.
2. Tavakoli lecture notes.
3. Lipka-Bartosik–Perarnau-Llobet–Brunner, arXiv:2308.15905.
4. Fux thesis on process tensor networks.
5. OQuPy, arXiv:2406.16650.
6. Rivas, arXiv:1910.01246.

6.5. What should not be done at the beginning

The following are common mistakes.

- Starting with encyclopedic textbooks on open quantum systems and tensor networks before understanding the baseline gate model.
- Treating non-Markovianity as a buzzword rather than specifying a bath model, memory scale, and observable.
- Discussing "heat" and "entropy production" beyond weak coupling before reading the strong-coupling thermodynamics literature carefully.
- Switching numerical frameworks too often before a first benchmark is obtained.

References

[1] P. P. Potts, *Introduction to Quantum Thermodynamics*, arXiv:1906.07439.

[2] A. Tavakoli, *Lecture notes: Quantum information in quantum thermal machines*, 2021, https://qthermo.ethz.ch/wp-content/uploads/2021/08/lecture_notes_les_diableretes.pdf.

[3] C. Latune, *Introduction to Quantum Thermodynamics*, lecture slides, https://indico.nithecs.ac.za/event/1/attachments/5/10/Lecture_1.pdf.

[4] P. Lipka-Bartosik, M. Perarnau-Llobet, and N. Brunner, *Thermodynamic Computing via Autonomous Quantum Thermal Machines*, arXiv:2308.15905.

[5] R. Finsterhölzl, M. Katzer, A. Knorr, and A. Carmele, *Using Matrix-Product States for Open Quantum Many-Body Systems: Efficient Algorithms for Markovian and Non-Markovian Time-Evolution*, *PRX Quantum* **1** (2020) 010701.

[6] G. E. Fux, *Process tensor networks for non-Markovian open quantum systems*, PhD thesis, University of St Andrews, 2022, `https://research-repository.st-andrews.ac.uk/handle/10023/26902`.

[7] *Process Tensor Approaches to Non-Markovian Quantum Dynamics*, perspective/review, `https://arxiv.org/html/2509.07661v1`.

[8] G. E. Fux, D. Kilda, B. W. Lovett, and E. M. Gauger, *Tensor network simulation of chains of non-Markovian open quantum systems*, arXiv:2201.05529.

[9] G. E. Fux, P. Fowler-Wright, J. Cerrillo, E. M. Gauger, and B. W. Lovett, *OQuPy: A Python package to efficiently simulate non-Markovian open quantum systems with process tensors*, arXiv:2406.16650.

[10] *MPSDynamics.jl: Tensor network simulations for finite-temperature environments*, arXiv:2406.07052.

[11] Á. Rivas, *Strong Coupling Thermodynamics of Open Quantum Systems*, arXiv:1910.01246.

[12] *Quantum Heat Current under Non-perturbative and Non-Markovian System-Reservoir Couplings*, arXiv:1609.08783.

[13] W.-M. Huang and W.-M. Zhang, *Strong Coupling Quantum Thermodynamics with Renormalized Hamiltonian and Temperature*, arXiv:2010.01828.

[14] *Non-Markovian thermal operations boosting the performance of heat engines*, *Physical Review E* **106** (2022) 014114.

[15] M. Rasola and M. Möttönen, *Autonomous Quantum Heat Engine Based on Non-Markovian Dynamics of an Optomechanical Hamiltonian*, arXiv:2403.18515.